

第七节 多元函数微分学的几何应用

一、空间曲线的切线与法平面

二、空间曲面的切平面与法线

三、小结

一、空间曲线的切线与法平面

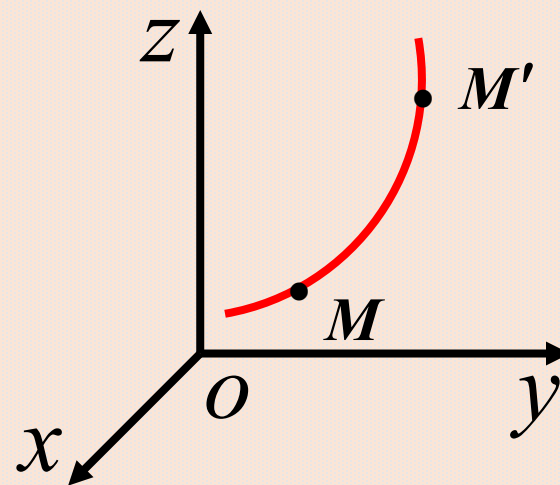
设空间曲线的方程
$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \\ z = \omega(t) \end{cases} \quad (1)$$

(1)式中的三个函数均可导.

设 $M(x_0, y_0, z_0)$, 对应于 $t = t_0$;

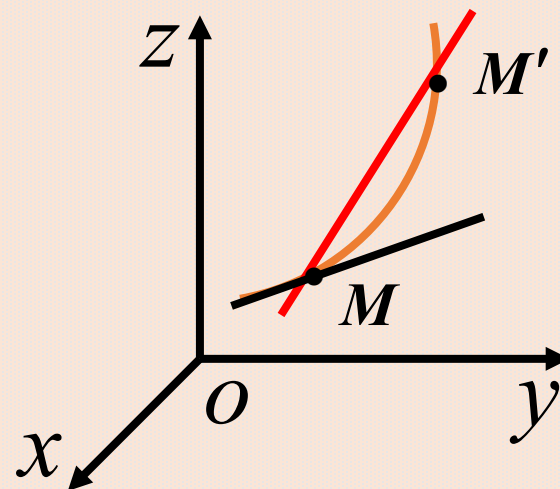
$M'(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$

对应于 $t = t_0 + \Delta t$.



割线 MM' 的方程为

$$\frac{x - x_0}{\Delta x} = \frac{y - y_0}{\Delta y} = \frac{z - z_0}{\Delta z}$$



考察割线趋近于极限位置——切线的过程

上式分母同除以 Δt ,

$$\frac{x - x_0}{\frac{\Delta x}{\Delta t}} = \frac{y - y_0}{\frac{\Delta y}{\Delta t}} = \frac{z - z_0}{\frac{\Delta z}{\Delta t}},$$

当 $M' \rightarrow M$, 即 $\Delta t \rightarrow 0$ 时,

曲线在 M 处的切线方程

$$\frac{x - x_0}{\varphi'(t_0)} = \frac{y - y_0}{\psi'(t_0)} = \frac{z - z_0}{\omega'(t_0)}.$$

切向量: 切线的方向向量称为曲线的切向量.

$$\vec{T} = \{\varphi'(t_0), \psi'(t_0), \omega'(t_0)\}$$

法平面: 过 M 点且与切线垂直的平面.

$$\varphi'(t_0)(x - x_0) + \psi'(t_0)(y - y_0) + \omega'(t_0)(z - z_0) = 0$$

例1 求曲线 $\Gamma : x = \int_0^t e^u \cos u du, y = 2 \sin t + \cos t, z = 1 + e^{3t}$ 在 $t = 0$ 处的切线和法平面方程.

解 当 $t = 0$ 时, $x = 0, y = 1, z = 2,$

$$x' = e^t \cos t, y' = 2 \cos t - \sin t, z' = 3e^{3t},$$

$$\Rightarrow x'(0) = 1, y'(0) = 2, z'(0) = 3,$$

切线方程 $\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{3},$

法平面方程 $x + 2(y-1) + 3(z-2) = 0,$

$$\text{即 } x + 2y + 3z - 8 = 0.$$

例2. 求圆柱螺旋线 $x = R \cos \varphi, y = R \sin \varphi, z = k\varphi$ 在 $\varphi = \frac{\pi}{2}$ 对应点处的切线方程和法平面方程.

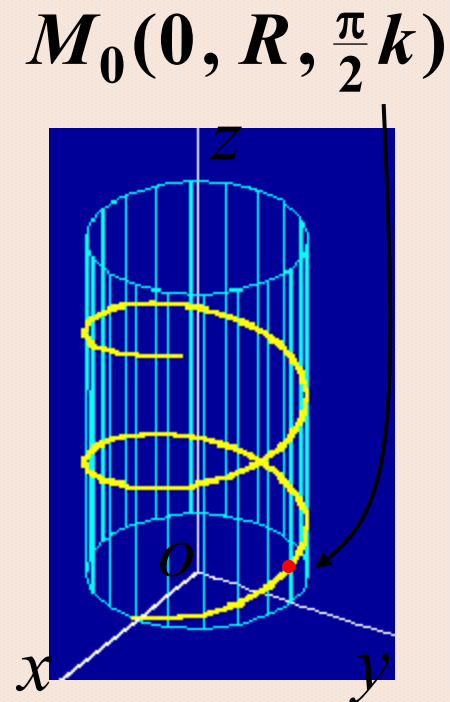
解: 由于 $x' = -R \sin \varphi, y' = R \cos \varphi, z' = k$, 当 $\varphi = \frac{\pi}{2}$ 时, 对应的切向量为 $\vec{T} = (-R, 0, k)$, 故

切线方程
$$\frac{x}{-R} = \frac{y - R}{0} = \frac{z - \frac{\pi}{2}k}{k}$$

即
$$\begin{cases} kx + Rz - \frac{\pi}{2}Rk = 0 \\ y - R = 0 \end{cases}$$

法平面方程 $-Rx + k(z - \frac{\pi}{2}k) = 0$

即 $Rx - kz + \frac{\pi}{2}k^2 = 0$



特殊地:

1. 空间曲线方程为 $\begin{cases} y = \varphi(x) \\ z = \psi(x) \end{cases},$

在 $M(x_0, y_0, z_0)$ 处,

切线方程为 $\frac{x - x_0}{1} = \frac{y - y_0}{\varphi'(x_0)} = \frac{z - z_0}{\psi'(x_0)},$

法平面方程为

$$(x - x_0) + \varphi'(x_0)(y - y_0) + \psi'(x_0)(z - z_0) = 0.$$

2.空间曲线方程为 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$,

切线方程为
$$\frac{x - x_0}{\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}_0} = \frac{y - y_0}{\begin{vmatrix} F'_z & F'_x \\ G'_z & G'_x \end{vmatrix}_0} = \frac{z - z_0}{\begin{vmatrix} F'_x & F'_y \\ G'_x & G'_y \end{vmatrix}_0},$$

法平面方程为

$$\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}_0 (x - x_0) + \begin{vmatrix} F'_z & F'_x \\ G'_z & G'_x \end{vmatrix}_0 (y - y_0) + \begin{vmatrix} F'_x & F'_y \\ G'_x & G'_y \end{vmatrix}_0 (z - z_0) = 0.$$

定理 设函数 $F(x, y, z), G(x, y, z)$ 满足:

① 在点 $P(x_0, y_0, z_0)$ 的某一邻域内具有连续偏导数;

② $F(x_0, y_0, z_0) = 0, G(x_0, y_0, z_0) = 0$;

③ $J \bigg|_P = \frac{\partial(F, G)}{\partial(y, z)} \bigg|_P \neq 0$

则方程组 $F(x, y, z) = 0, G(x, y, z) = 0$ 在点 x_0 的某一邻域内可**唯一**确定一组满足条件 $y_0 = y(x_0), z_0 = z(x_0)$

的**单值连续函数** $y = y(x), z = z(x)$, 且有偏导数公式:

$$y'(x) = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, z)} = -\frac{\begin{vmatrix} F'_x & F'_z \\ G'_y & G'_z \end{vmatrix}}{\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}}, \quad z'(x) = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, x)} = -\frac{\begin{vmatrix} F'_y & F'_x \\ G'_y & G'_x \end{vmatrix}}{\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}}.$$

设方程组 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 有隐函数组 $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$, 则

$$\begin{cases} F(x, y(x), z(x)) \equiv 0 \\ G(x, y(x), z(x)) \equiv 0 \end{cases}$$

两边对 x 求导得 $\begin{cases} F'_x + F'_y \cdot y'(x) + F'_z \cdot z'(x) = 0 \\ G'_x + G'_y \cdot y'(x) + G'_z \cdot z'(x) = 0 \end{cases}$

这是关于 $y'(x)$, $z'(x)$ 的线性方程组, 在点 P 的某邻域内

系数行列式 $J = \begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix} \neq 0$, 故解得

$$y'(x) = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, z)}, \quad z'(x) = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, x)}.$$

例3 求曲线 $x^2 + y^2 + z^2 = 6$, $x + y + z = 0$ 在点 $(1, -2, 1)$ 处的切线及法平面方程.

解1 直接利用公式;

解2 将所给方程的两边对 x 求导并移项, 得

$$\begin{cases} y \frac{dy}{dx} + z \frac{dz}{dx} = -x \\ \frac{dy}{dx} + \frac{dz}{dx} = -1 \end{cases} \Rightarrow \begin{cases} \frac{dy}{dx} = \frac{z-x}{y-z}, \\ \frac{dz}{dx} = \frac{x-y}{y-z}, \end{cases}$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(1,-2,1)} = 0, \quad \left. \frac{dz}{dx} \right|_{(1,-2,1)} = -1,$$

由此得切向量 $\vec{T} = \{1, 0, -1\}$,

所求切线方程为 $\frac{x-1}{1} = \frac{y+2}{0} = \frac{z-1}{-1},$

法平面方程为 $(x-1) + 0 \cdot (y+2) - (z-1) = 0,$

$$\Rightarrow x - z = 0$$

二、曲面的切平面与法线

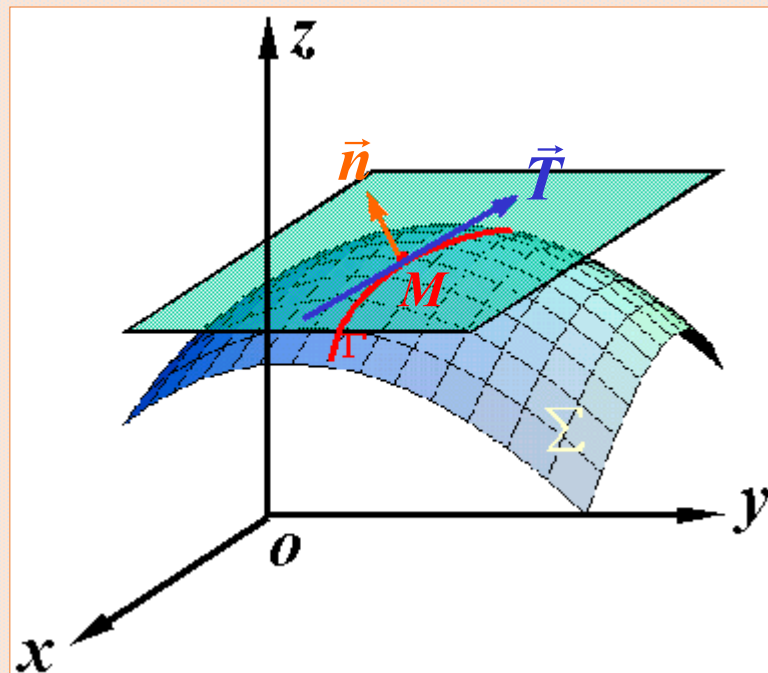
设曲面方程为

$$F(x, y, z) = 0$$

在曲面上任取一条通过点 M 的曲线

$$\Gamma: \begin{cases} x = \varphi(t) \\ y = \psi(t), \\ z = \omega(t) \end{cases}$$

曲线在 M 处的切向量 $\vec{T} = \{\varphi'(t_0), \psi'(t_0), \omega'(t_0)\}$,



$\because \Gamma : x = \varphi(t), y = \psi(t), z = \omega(t)$ 在 Σ 上,

$$\therefore F(\varphi(t), \psi(t), \omega(t)) \equiv 0$$

两边在 $t = t_0$ 处求导, 注意 $t = t_0$ 对应点 M ,

得

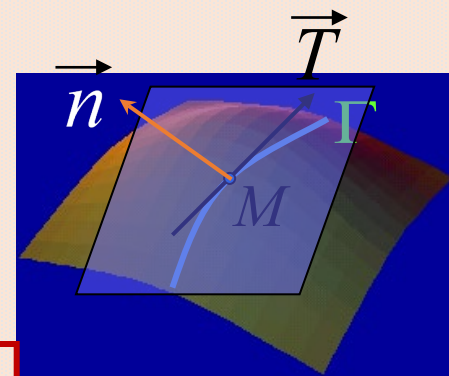
$$F'_x(x_0, y_0, z_0) \varphi'(t_0) + F'_y(x_0, y_0, z_0) \psi'(t_0) + F'_z(x_0, y_0, z_0) \omega'(t_0) = 0$$

$$\left| \begin{array}{l} \text{令 } \vec{T} = (\varphi'(t_0), \psi'(t_0), \omega'(t_0)) \end{array} \right.$$

$$\left| \begin{array}{l} \vec{n} = (F'_x(x_0, y_0, z_0), F'_y(x_0, y_0, z_0), F'_z(x_0, y_0, z_0)) \end{array} \right.$$

切向量 $\vec{T} \perp \vec{n}$

由于曲线 Γ 的任意性, 表明这些切线都在以 $\vec{n}$ 为法向量的平面上, 称此平面为该点的切平面, 从而切平面存在.



曲面 Σ 在点 M 的法向量

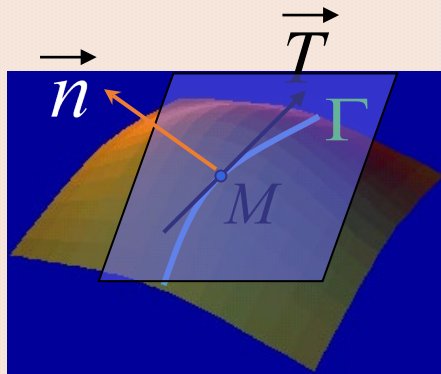
$$\vec{n} = (F'_x(x_0, y_0, z_0), F'_y(x_0, y_0, z_0), F'_z(x_0, y_0, z_0))$$

切平面方程

$$F'_x(x_0, y_0, z_0)(x - x_0) + F'_y(x_0, y_0, z_0)(y - y_0) + F'_z(x_0, y_0, z_0)(z - z_0) = 0$$

法线方程

$$\frac{x - x_0}{F'_x(x_0, y_0, z_0)} = \frac{y - y_0}{F'_y(x_0, y_0, z_0)} = \frac{z - z_0}{F'_z(x_0, y_0, z_0)}$$



特别, 当光滑曲面 Σ 的方程为显式 $z = f(x, y)$ 时, 令

$$F(x, y, z) = f(x, y) - z$$

则在点 (x, y, z) , $F'_x = f'_x$, $F'_y = f'_y$, $F'_z = -1$

故当函数 $f(x, y)$ 在点 (x_0, y_0) 有连续偏导数时, 曲面
 Σ 在点 (x_0, y_0, z_0) 有

切平面方程

$$z - z_0 = f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$$

法线方程

$$\frac{x - x_0}{f'_x(x_0, y_0)} = \frac{y - y_0}{f'_y(x_0, y_0)} = \frac{z - z_0}{-1}$$

用 α, β, γ 表示法向量的方向角, 并假定法向量方向向上, 则 γ 为锐角.

法向量 $\vec{n} = (-f'_x(x_0, y_0), -f'_y(x_0, y_0), 1)$

将 $f'_x(x_0, y_0), f'_y(x_0, y_0)$ 分别记为 f'_x, f'_y , 则

法向量的方向余弦:

$$\cos \alpha = \frac{-f'_x}{\sqrt{1 + f'^2_x + f'^2_y}}, \quad \cos \beta = \frac{-f'_y}{\sqrt{1 + f'^2_x + f'^2_y}},$$
$$\cos \gamma = \frac{1}{\sqrt{1 + f'^2_x + f'^2_y}}$$

全微分的几何意义

因为曲面在 M 处的切平面方程为

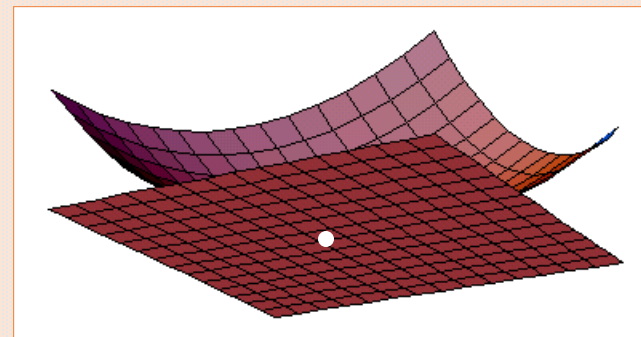
$$z - z_0 = f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$$

切平面上点的
竖坐标的增量

函数 $z = f(x, y)$ 在点 (x_0, y_0) 的全微分

$z = f(x, y)$ 在 (x_0, y_0) 的全微分，表示曲面 $z = f(x, y)$ 在点 (x_0, y_0, z_0) 处的切平面上的点的竖坐标的增量.

例3 求旋转抛物面 $z = x^2 + y^2 - 1$ 在点 $(2, 1, 4)$ 处的切平面及法线方程.



解 $f(x, y) = x^2 + y^2 - 1,$

$$\vec{n}|_{(2,1,4)} = \{2x, 2y, -1\}|_{(2,1,4)} = \{4, 2, -1\},$$

切平面方程为 $4(x - 2) + 2(y - 1) - (z - 4) = 0,$

$$\Rightarrow 4x + 2y - z - 6 = 0,$$

法线方程为 $\frac{x - 2}{4} = \frac{y - 1}{2} = \frac{z - 4}{-1}.$

例4 求曲面 $z - e^z + 2xy = 3$ 在点 $(1, 2, 0)$ 处的切平面及法线方程.

解 令 $F(x, y, z) = z - e^z + 2xy - 3,$

$$F'_x|_{(1,2,0)} = 2y|_{(1,2,0)} = 4, \quad F'_y|_{(1,2,0)} = 2x|_{(1,2,0)} = 2,$$

$$F'_z|_{(1,2,0)} = 1 - e^z|_{(1,2,0)} = 0,$$

切平面方程 $4(x-1) + 2(y-2) + 0 \cdot (z-0) = 0,$

$$\Rightarrow 2x + y - 4 = 0,$$

法线方程 $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-0}{0}.$

例 5 求曲面 $x^2 + 2y^2 + 3z^2 = 21$ 平行于平面 $x + 4y + 6z = 0$ 的各切平面方程.

解 设 (x_0, y_0, z_0) 为曲面上的切点,
切平面方程为

$$2x_0(x - x_0) + 4y_0(y - y_0) + 6z_0(z - z_0) = 0$$

依题意, 切平面方程平行于已知平面, 得

$$\frac{2x_0}{1} = \frac{4y_0}{4} = \frac{6z_0}{6}, \Rightarrow 2x_0 = y_0 = z_0.$$

因为 (x_0, y_0, z_0) 是曲面上的切点,

满足方程 $\therefore x_0 = \pm 1$,

所求切点为 $(1, 2, 2), (-1, -2, -2)$,

切平面方程(1)

$$2(x-1) + 8(y-2) + 12(z-2) = 0$$

$$\Rightarrow x + 4y + 6z = 21$$

切平面方程(2)

$$-2(x+1) - 8(y+2) - 12(z+2) = 0$$

$$\Rightarrow x + 4y + 6z = -21$$

例4. 确定正数 σ 使曲面 $x y z = \sigma$ 与球面 $x^2 + y^2 + z^2 = a^2$ 在点 $M(x_0, y_0, z_0)$ 相切.

解: 二曲面在 M 点的法向量分别为

$$\vec{n}_1 = (y_0 z_0, x_0 z_0, x_0 y_0), \quad \vec{n}_2 = (x_0, y_0, z_0)$$

二曲面在点 M 相切, 故 $\vec{n}_1 \parallel \vec{n}_2$, 因此有

$$\frac{x_0 y_0 z_0}{x_0^2} = \frac{x_0 y_0 z_0}{y_0^2} = \frac{x_0 y_0 z_0}{z_0^2}$$

$$\therefore x_0^2 = y_0^2 = z_0^2$$

又点 M 在球面上, 故 $x_0^2 = y_0^2 = z_0^2 = \frac{a^2}{3}$

于是有 $\sigma = x_0 y_0 z_0 = \frac{a^3}{3\sqrt{3}}$

例 5 设 $\vec{n}$ 是曲面 $2x^2 + 3y^2 + z^2 = 6$ 在点 $P(1,1,1)$ 处的指向外侧的法向量, 求函数 $u = \frac{1}{z}(6x^2 + 8y^2)^{\frac{1}{2}}$ 在此处沿方向 $\vec{n}$ 的方向导数.

解 令 $F(x, y, z) = 2x^2 + 3y^2 + z^2 - 6,$

$$F'_x|_P = 4x|_P = 4, \quad F'_y|_P = 6y|_P = 6, \quad F'_z|_P = 2z|_P = 2,$$

$$\text{故 } \vec{n} = \{F'_x, F'_y, F'_z\} = \{4, 6, 2\},$$

$$|\vec{n}| = \sqrt{4^2 + 6^2 + 2^2} = 2\sqrt{14}, \quad \text{方向余弦为}$$

$$\cos \alpha = \frac{2}{\sqrt{14}}, \quad \cos \beta = \frac{3}{\sqrt{14}}, \quad \cos \gamma = \frac{1}{\sqrt{14}}.$$

$$\left. \frac{\partial u}{\partial x} \right|_P = \left. \frac{6x}{z\sqrt{6x^2 + 8y^2}} \right|_P = \frac{6}{\sqrt{14}};$$

$$\left. \frac{\partial u}{\partial y} \right|_P = \left. \frac{8y}{z\sqrt{6x^2 + 8y^2}} \right|_P = \frac{8}{\sqrt{14}};$$

$$\left. \frac{\partial u}{\partial z} \right|_P = \left. -\frac{\sqrt{6x^2 + 8y^2}}{z^2} \right|_P = -\sqrt{14}.$$

$$\text{故 } \left. \frac{\partial u}{\partial \vec{n}} \right|_P = \left(\frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma \right) \Big|_P = \frac{11}{7}.$$

内容小结

1. 空间曲线的切线与法平面

1) 参数式情况. 空间光滑曲线 $\Gamma : \begin{cases} x = \varphi(t) \\ y = \psi(t) \\ z = \omega(t) \end{cases}$

切向量 $\vec{T} = (\varphi'(t_0), \psi'(t_0), \omega'(t_0))$

切线方程 $\frac{x - x_0}{\varphi'(t_0)} = \frac{y - y_0}{\psi'(t_0)} = \frac{z - z_0}{\omega'(t_0)}$

法平面方程

$$\varphi'(t_0)(x - x_0) + \psi'(t_0)(y - y_0) + \omega'(t_0)(z - z_0) = 0$$

2) 一般式情况. 空间光滑曲线 $\Gamma: \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$

切向量 $\vec{T} = \left(\frac{\partial(F, G)}{\partial(y, z)} \Big|_M, \frac{\partial(F, G)}{\partial(z, x)} \Big|_M, \frac{\partial(F, G)}{\partial(x, y)} \Big|_M \right)$

切线方程 $\frac{x - x_0}{\frac{\partial(F, G)}{\partial(y, z)} \Big|_M} = \frac{y - y_0}{\frac{\partial(F, G)}{\partial(z, x)} \Big|_M} = \frac{z - z_0}{\frac{\partial(F, G)}{\partial(x, y)} \Big|_M}$

法平面方程 $\frac{\partial(F, G)}{\partial(y, z)} \Big|_M (x - x_0) + \frac{\partial(F, G)}{\partial(z, x)} \Big|_M (y - y_0) + \frac{\partial(F, G)}{\partial(x, y)} \Big|_M (z - z_0) = 0$

2. 曲面的切平面与法线

1) 隐式情况 . 空间光滑曲面 $\Sigma : F(x, y, z) = 0$

曲面 Σ 在点 $M(x_0, y_0, z_0)$ 的**法向量**

$$\vec{n} = (F'_x(x_0, y_0, z_0), F'_y(x_0, y_0, z_0), F'_z(x_0, y_0, z_0))$$

切平面方程

$$F'_x(x_0, y_0, z_0)(x - x_0) + F'_y(x_0, y_0, z_0)(y - y_0) + F'_z(x_0, y_0, z_0)(z - z_0) = 0$$

法线方程

$$\frac{x - x_0}{F'_x(x_0, y_0, z_0)} = \frac{y - y_0}{F'_y(x_0, y_0, z_0)} = \frac{z - z_0}{F'_z(x_0, y_0, z_0)}$$

2) 显式情况. 空间光滑曲面 $\Sigma : z = f(x, y)$

法向量 $\vec{n} = (-f'_x, -f'_y, 1)$

法线的方向余弦

$$\cos \alpha = \frac{-f'_x}{\sqrt{1 + f'^2_x + f'^2_y}}, \quad \cos \beta = \frac{-f'_y}{\sqrt{1 + f'^2_x + f'^2_y}},$$

$$\cos \gamma = \frac{1}{\sqrt{1 + f'^2_x + f'^2_y}}$$

切平面方程

$$z - z_0 = f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$$

法线方程

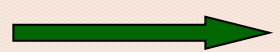
$$\frac{x - x_0}{f'_x(x_0, y_0)} = \frac{y - y_0}{f'_y(x_0, y_0)} = \frac{z - z_0}{-1}$$

思考与练习

1. 如果平面 $3x + \lambda y - 3z + 16 = 0$ 与椭球面 $3x^2 + y^2 + z^2 = 16$ 相切, 求 λ .

提示: 设切点为 $M(x_0, y_0, z_0)$, 则

$$\begin{cases} \frac{6x_0}{3} = \frac{2y_0}{\lambda} = \frac{2z_0}{-3} & \text{(二法向量平行)} \\ 3x_0 + \lambda y_0 - 3z_0 + 16 = 0 & \text{(切点在平面上)} \\ 3x_0^2 + y_0^2 + z_0^2 = 16 & \text{(切点在椭球面上)} \end{cases}$$

 $\lambda = \pm 2$

2. 设 $f(u)$ 可微, 证明 曲面 $z = xf(\frac{y}{x})$ 上任一点处的切平面都通过原点.

提示: 在曲面上任意取一点 $M(x_0, y_0, z_0)$, 则通过此点的切平面为

$$z - z_0 = \left. \frac{\partial z}{\partial x} \right|_M (x - x_0) + \left. \frac{\partial z}{\partial y} \right|_M (y - y_0)$$

证明原点坐标满足上述方程 .

3. 证明曲面 $F(x-my, z-ny)=0$ 的所有切平面恒与定直线平行, 其中 $F(u,v)$ 可微.

证: 曲面上任一点的法向量

$$\vec{n} = (F'_1, F'_1 \cdot (-m) + F'_2 \cdot (-n), F'_2)$$

取定直线的方向向量为 $\vec{l} = (m, 1, n)$ (定向量)

则 $\vec{l} \cdot \vec{n} = 0$, 故结论成立.

4. 求曲线 $\begin{cases} x^2 + y^2 + z^2 - 3x = 0 \\ 2x - 3y + 5z - 4 = 0 \end{cases}$ 在点(1,1,1) 的切线
与法平面.

解: 点 (1,1,1) 处两曲面的法向量为

$$\vec{n}_1 = (2x - 3, 2y, 2z) \Big|_{(1,1,1)} = (-1, 2, 2)$$

$$\vec{n}_2 = (2, -3, 5)$$

因此切线的方向向量为 $\vec{l} = \vec{n}_1 \times \vec{n}_2 = (16, 9, -1)$

由此得切线: $\frac{x-1}{16} = \frac{y-1}{9} = \frac{z-1}{-1}$

法平面: $16(x-1) + 9(y-1) - (z-1) = 0$

即 $16x + 9y - z - 24 = 0$